

# Akshit Gupta

Master of Operational Research | akshitgupta.du.or.27@gmail.com | 8130020286

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## Education

|                                                                                                                 |             |
|-----------------------------------------------------------------------------------------------------------------|-------------|
| <b>Department of Operational Research (Hansraj College, University of Delhi)</b> Master of Operational Research | 2025 – 2027 |
| • SGPA(1 <sup>st</sup> year) : 7.2                                                                              |             |
| <b>Ramanujan College, University of Delhi</b> Bvoc. Software Development(IT/ES)                                 | 2021 – 2024 |
| • CGPA: 8.033                                                                                                   |             |
| <b>Delhi Public School, Bulandshahr,Uttar Pradesh</b> Boards-XII (PCM)                                          | 2020 – 2021 |
| • 87.6%                                                                                                         |             |
| <b>Delhi Public School, Bulandshahr,Uttar Pradesh</b> Boards-X                                                  | 2018 – 2019 |
| • 87.1%                                                                                                         |             |

## Projects

### PRISM – Predictive & Research-based Insurance Statistical Modeling

*Python, Statistics, Machine Learning, GLM, XGBoost, Causal Inference*

- Built an **end-to-end insurance pricing system modeling** claim frequency and severity using **Poisson, Negative Binomial, and Gamma GLMs**, combining them into a pure premium pricing framework.
- **Benchmarked interpretable statistical models against XGBoost**, achieving a **3% performance improvement** while selecting the final model based on explainability and stability.
- **Applied causal inference** using propensity score weighting to identify true risk drivers and justify pricing factors.
- Translated model outputs into **business-ready risk segmentation** enabling fair and profitable pricing decisions.

Link: <https://github.com/akshit20421/prism-insurance-risk-modeling>

### Credit Risk Analytics Dashboard | Power BI, DAX, Power Query

*Power BI, DAX, Power Query, Risk Analytics*

- Developed a 4-page Credit Risk Analytics Dashboard **analyzing 255K+ loan records** for **portfolio monitoring and borrower risk assessment**.
- Built **interactive KPI reporting, borrower segmentation, default trend analysis, and Expected Credit Loss (ECL)** monitoring using Power BI and DAX.
- Analyzed **risk drivers across Credit Score, DTI, Income, Employment Type, Loan Purpose, and Interest Rate** attributes to identify high-risk borrower segments.
- Designed **drill-down dashboards with slicers and dynamic filtering** capabilities to support data-driven lending and risk management decisions.

Link: <https://github.com/akshit20421/credit-risk-analytics-dashboard>

### Fair Pricing Engine | Explainable & Calibrated Insurance ML System

*Python, XGBoost, Statsmodels (Gamma GLM), SHAP, Scikit-learn*

- Designed an **end-to-end insurance premium pricing system** integrating a Gamma **GLM baseline and advanced XGBoost models**.
- **Achieved ~40% RMSE reduction over the GLM benchmark** on identical test splits using enriched medical and lifestyle features.
- Performed **decile-level calibration analysis** to validate pricing stability across risk segments.
- **Implemented SHAP-based global and local explanations** for model transparency.

Link: <https://github.com/akshit20421/fair-explainable-pricing-engine>

## Positions of Responsibility

### Microsoft Learn Student Ambassador (MLSA)

- Represented Microsoft as a student ambassador, engaging peers with workshops, events, and tech community activities.

### Technical Director, Ramanujan Consulting Club, Ramanujan College, University of Delhi (Dec'22-May'23)

- Led a 10+ member tech team, streamlined development workflows by 40%, and delivered client-focused digital solutions in collaboration with consulting teams.

## Technical Skills

**Tools/ Libraries:** Python, Pandas, NumPy, Statsmodels, Matplotlib, Git/GitHub, Sql,Power BI, DAX, Power Query, Excel

## Certifications :

Aplha Microsoft Student Ambassador 2024 - **Microsoft** , Python – **Kaggle**,DSA JS- **freecodecamp**